

#### 4.2.4 Resistivity

| Learning outcomes                                                                             |                                                                                              | Additional guidance |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> |                                                                                              |                     |
| (a)                                                                                           | (i) resistivity of a material; the equation<br>$R = \frac{\rho L}{A}$                        | <b>PAG3</b>         |
|                                                                                               | (ii) techniques and procedures used to determine the resistivity of a metal.                 |                     |
| (b)                                                                                           | the variation of resistivity of metals and semiconductors with temperature                   | HSW2                |
| (c)                                                                                           | negative temperature coefficient (NTC) thermistor; variation of resistance with temperature. | HSW5                |